

1 In this experiment, you will investigate oscillations.

You have been provided with a double spring and two single springs.

- (a) (i)**
- Slide the double spring and the two single springs onto the longer wooden rod.
 - Set up the apparatus as shown in Fig. 1.1.

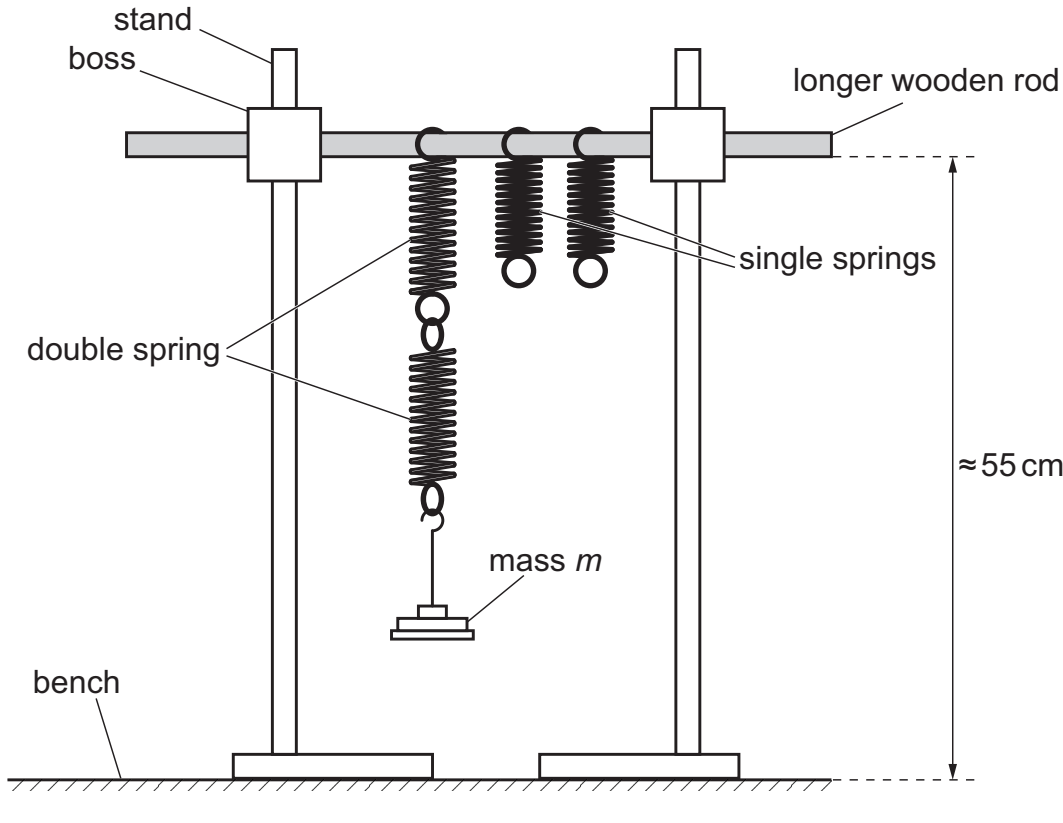


Fig. 1.1

- Use the bosses to fix the wooden rod approximately 55 cm above the bench and parallel to the bench.
- Hang a total mass of 270 g from the double spring.
- The mass hanging from the double spring is m .

Record m .

$$m = \dots\dots\dots [1]$$

- (ii)**
- Gently pull the mass down through a short distance. When released, the mass will oscillate.

- Take measurements to determine the period T_1 of the oscillations.

$$T_1 = \dots\dots\dots$$

- Remove the mass from the double spring.

[2]

- (b)**
- Using the shorter wooden rod, hang mass m as shown in Fig. 1.2.

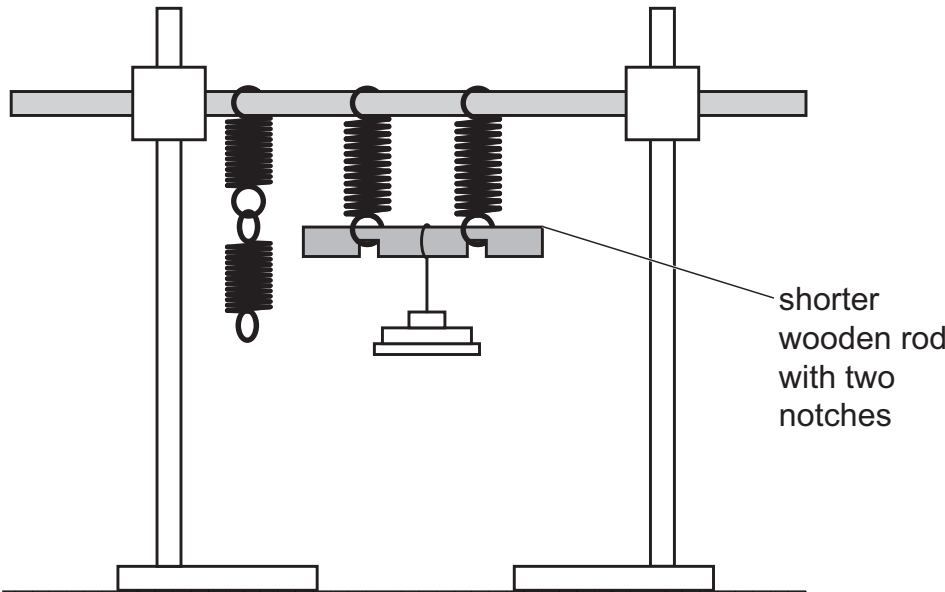


Fig. 1.2

- Gently** place the hook of the mass hanger near the centre of the shorter rod.
- Carefully** adjust the position of the mass hanger until the shorter rod is approximately parallel to the bench, as shown in Fig. 1.2.
- Gently pull the mass down through a short distance. When released, the mass will oscillate.
- Take measurements to determine the period T_2 of the oscillations.

$$T_2 = \dots\dots\dots$$

- Remove the mass.

[1]

- (c)** Vary m . For each value of m , determine T_1 and T_2 . Repeat until you have five sets of values of m , T_1 and T_2 . Do **not** use values of m less than 200 g.

Record your results in a table. Include values of $\sqrt{T_1}$ and $\sqrt{T_2}$ in your table.

[8]

- (d) (i)** Plot a graph of $\sqrt{T_2}$ on the y -axis against $\sqrt{T_1}$ on the x -axis.

[3]

- (ii)** Draw the straight line of best fit.

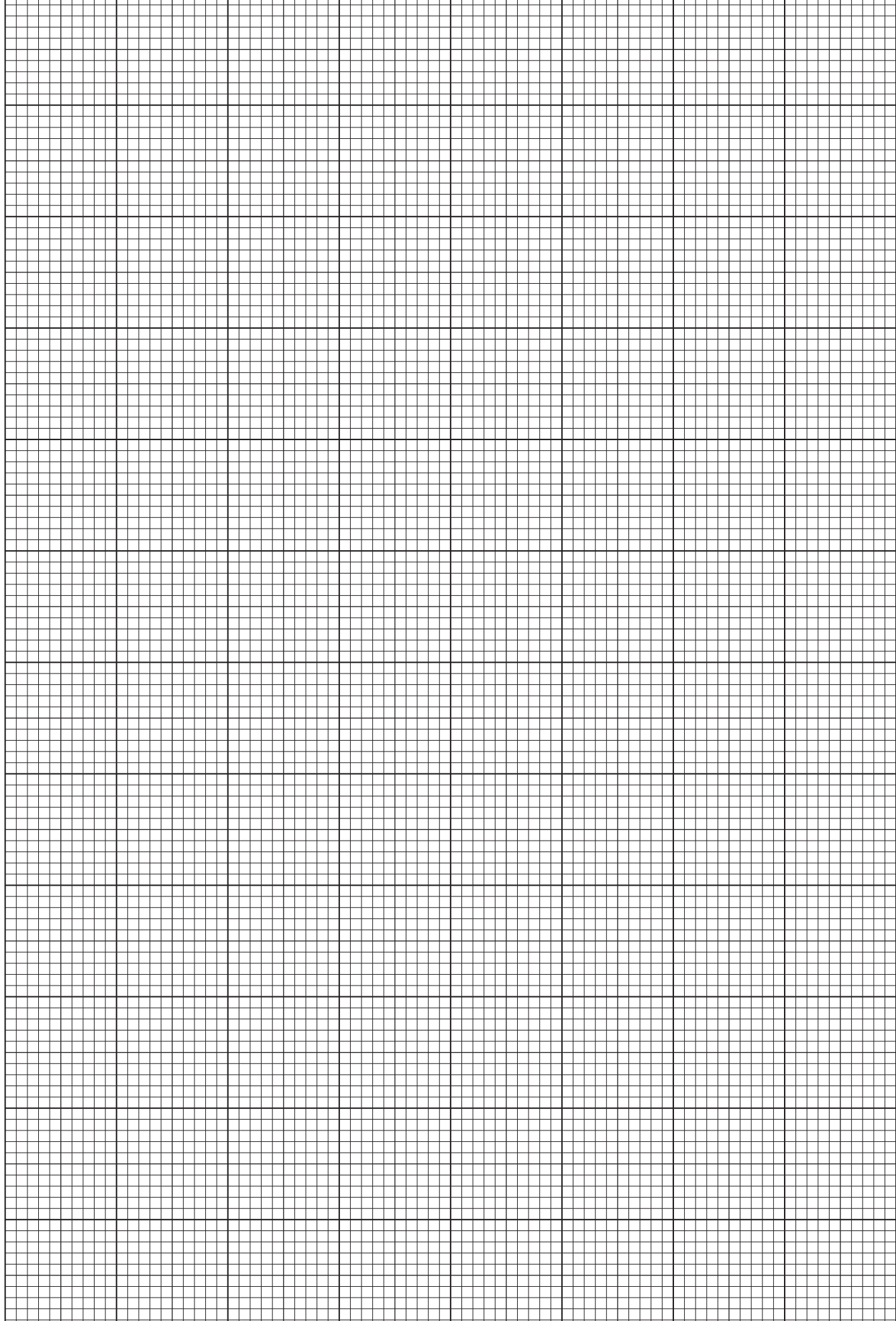
[1]

- (iii)** Determine the gradient and y -intercept of this line.

$$\text{gradient} = \dots\dots\dots$$

$$y\text{-intercept} = \dots\dots\dots$$

[2]



- (e)** It is suggested that the quantities $\sqrt{T_2}$ and $\sqrt{T_1}$ are related by the equation

$$\sqrt{T_2} = P\sqrt{T_1} + Q$$

where P and Q are constants.

Using your answers in **(d)(iii)**, determine the values of P and Q .
Give appropriate units.

$$P = \dots\dots\dots$$

$$Q = \dots\dots\dots$$

[2]

[Total: 20]

You may not need to use all of the materials provided.